

US Military Intervention and Civilian Casualty Patterns

Project Proposal Report

Data Science | Spring 2026

1. Problem Statement and Importance

The United States directly killed a minimum of 22,679 civilians in airstrikes carried out since the start of the War on Terror in 2001. Yet the US military's own acknowledgment for the ISIS campaign alone was approximately 1,500 civilian deaths while independent monitors have estimated between 8,000 and 13,000. These patterns of underreporting and the conditions under which civilian harm escalates remain largely unexamined at scale.

The core question this project asks is whether the patterns in US military strikes, whether the locations, types, contexts, and scales, can be used to predict civilian casualty counts that the US military itself systematically underreports. This is not isolated to one campaign either. Across Iraq, Syria, Afghanistan, and Yemen there have been thousands of civilian casualties. Airwars documents civilian deaths across these conflicts with maximum estimates reaching nearly 48,000, many of them named and documented, yet never acknowledged by the US military. The emergence of large-scale independent conflict databases like Airwars and UCDP now makes systematic analysis of these patterns feasible. If we can model these patterns, then global organisations—the International Criminal Court (ICC), journalists, and humanitarian organisations—can use this to pursue accountability that official military reporting actively obscures.

The populations most directly affected by US military strikes—civilians in Iraq, Syria, Afghanistan, and Yemen—have been largely ignored and deprived of accountability for the afflictions brought on them by US intervention. Humanitarian organisations can use predictive models of civilian harm to prepare and position aid, medical resources, displacement support, and psychological resources before or immediately after a strike rather than reacting weeks later. For legal bodies like the ICC, a predictive model offers independent and data-driven estimates of anticipated civilian casualties. These estimates can serve as independent evidence to challenge strikes before they occur, or to establish disregard for civilian life after the fact.

2. Literature Review

2.1 Machine Learning and Conflict Prediction: A Use Case

Perry, C. (2013). *Stability: International Journal of Security & Development*, 2(3), 56, pp. 1–18.

<https://doi.org/10.5334/sta.cr>

Problem. The author is trying to solve whether it is feasible to use a combination of open source data and national statistics to find indicators of a vulnerability to a violent event, whether aggregation and modelling can be parallelised to handle larger data-sets in real time, and if modern computing techniques can be used to predict conflict and feed into 3rd and 4th generation early warning systems.

Methodology. The study introduces conflict data drawn from ACLED, covering over 50 countries from 1997 to 2012, aggregated to district level using administrative boundaries from Columbia's GRUMP dataset, alongside socioeconomic indicators including GDP, population, ethnic composition, and governance data. The author uses

this data to test the efficacy of two models—Naive Bayes and Random Forest – in their abilities to predict the occurrence and intensity of armed conflict at the district level.

Results and Findings. Model specification was an important factor in the efficacy of solving the problem—Naive Bayes using only prior battles as a predictor achieved 18.3% conflict event prediction accuracy, whereas default Random Forest achieved 58.5%. Full models incorporating socioeconomic and geographic features also outperformed those relying on prior conflict history alone, suggesting that contextual indicators carry meaningful predictive power beyond historical violence.

2.2 The Promise of Machine Learning in Violent Conflict Forecasting

Murphy, M., Sharpe, E., & Huang, K. (2024). *Data & Policy*, 6, e35. <https://doi.org/10.1017/dap.2024.27>

Problem. The authors are attempting to examine why the international community continuously fails at forecasting violent conflict, especially with unexpected outbreaks like Ethiopia's 2020 Tigray crisis. They ask whether machine learning, combined with expanding open-source data, can meaningfully improve the accuracy of forecasting. Their central argument is that machine learning's promise depends on solving two bottlenecks: poor data quality and lack of model interpretability, which tends to prevent policymakers from trusting predictions.

Methodology. With this paper being a commentary rather than an empirical study, the authors are focused on reviewing the evolution of conflict forecasting and survey the major data sources and modeling approaches. Their focus is on identifying the technical and policy conditions under which machine learning can responsibly support conflict prediction.

Results and Findings. The authors conclude that machine learning-based forecasting holds promise but is only useful if the data quality improves and models become more interpretable. They highlight key findings from prior research: conflict recurrence is about 30% immediately after a conflict ends but falls to 0.5% after ten years of peace, and major datasets can diverge dramatically, for example in Mexico in 2021 where ACLED reported 6,739 civilian deaths while UCDP-GED reported 28. The authors argue that forecasting should be used heuristically to flag emerging risks rather than as a deterministic tool.

2.3 Prevention Is Better Than Cure: Machine Learning Approach to Conflict Prediction in Sub-Saharan Africa

Musumba, M., Fatema, N., & Kibriya, S. (2021). *Sustainability*, 13(13), 7366. <https://doi.org/10.3390/su13137366>

Problem. The authors are trying to determine whether machine learning classification algorithms can more accurately predict civil conflict than traditional logistic regression, using fine-grained sub-national data across sub-Saharan Africa. They aim to assess whether addressing class imbalance, selecting appropriate performance metrics, and comparing multiple algorithms can produce more reliable early-warning predictions that help policymakers prevent conflict rather than respond after the fact.

Methodology. The study uses ACLED conflict data 1997-2016 combined with geographic, socioeconomic, political, and agricultural variables at a 0.5°x0.5° grid resolution, producing 5,928 observations across 48 African countries. Five models are compared based on accuracy, precision, recall, and F1 score: logistic regression, random forest, gradient boosting, support vector machines, and multilayer perceptron. As only 7.9% of grid cells

experience conflict, the authors test several resampling strategies and ultimately apply SMOTE to correct class imbalance.

Results and Findings. Gradient boosting achieves the strongest baseline performance with 0.94 accuracy, 0.423 recall, and 0.543 F1 score. Whereas logistic regression and multilayer perceptron perform moderately well. After applying SMOTE, the recall increases substantially across all of the models; gradient boosting becomes the best model when recall is prioritised, whereas multilayer perceptron becomes the best when precision or F1 score is the chosen metric. The authors conclude that model choice depends on policy goals: preventing missed conflicts favours high-recall models like gradient boosting, while balanced performance favours multilayer perceptron. Rankings of variable importance highlight past conflict, night-light density, proximity to rivers, industrial metals, and agricultural production systems as key predictors.

3. Data

This project utilises four datasets. The primary one is UCDP-GED as it has global coverage spanning from 1989-2024, is well-structured with well-defined features within the official codebook, and has fatality estimates – an important metric for the problem statement. It will be supplemented by Airwars, ACLED, and IBC (Iraq Body Count). Airwars provides independent estimates that expose the gap between US official counts and real counts, ACLED adds broader context of the conflicts such as event type and actor behaviour, and IBC offers a second count specifically for Iraq to compare against. All data sets are publicly available at the following sources:

- UCDP-GED: <https://ucdp.uu.se/downloads>
- Airwars: <https://airwars.org/belligerent/us-forces/>
- ACLED: <https://acleddata.com/data-export-tool>
- IBC: <https://www.iraqbodycount.org/database>

UCDP-GED contains over 300,000 globally georeferenced events across the full period of coverage. After filtering UCDP-GED to events where the US or a US-led coalition is recorded as the perpetrating party across Syria, Iraq, Afghanistan, and Yemen, the working dataset to be used spans from 2001 to 2024. This subset will be determined during the preprocessing phase, though US-involved events in the four mentioned countries represent a substantial portion of Middle East and Central Asia entries. The temporal resolution is daily and the spatial resolution goes down to the level of individual towns or villages with full geocoding. Some known limitations include conservative estimates of fatalities and potential underreporting in conflict zones with limited media access.

Input variables:

From UCDP-GED:

- **country:** country in which the event takes place (categorical)
- **year:** year of the event (integer)
- **month:** extracted from *date_start*, captures seasonal patterns (integer)
- **type_of_violence:** state-based, non-state, or one-sided (categorical)
- **region:** broader geographic context (categorical)
- **latitude/longitude:** precise location, useful for spatial clustering (continuous)
- **side_b:** who the US was fighting against, i.e. Taliban, Isis, etc (categorical)

- **best**: total fatality estimate, strong predictor of civilian harm (integer)
- **event_clarity**: how precisely the event was documented, 1 or 2 (categorical)

From ACLED:

- **event_type**: airstrike, ground assault, etc (categorical)
- **sub_event_type**: more granular action classification (categorical)
- **interaction**: interaction code between actor types, i.e. military vs military or military forces vs civilians (categorical)

From Airwars:

- Civilian casualty allegation confidence rating – high, medium, low (categorical)
- Minimum and maximum independent casualty estimates (integer)

From IBC (validation only):

- Monthly maximum recorded civilian deaths in Iraq – used for validation against model’s predictions, not as a training feature (integer)

Derived feature(s):

- **administration**: US presidential administration during the event, derived from **year** (categorical)

Output Variable:

- **deaths_civilians**: best estimate of civilian deaths per event (integer, regression target)

4. Methodology and Evaluation Plan:

The dataset will first be filtered to only include events where the perpetrators are the US or US-led coalitions. Unnecessary features such as **relid** (deprecated feature) and **conflict_new_id** (unique identification code for each event), will be removed to reduce noise and retain only features that are relevant to predicting civilian casualties. Rows with any missing values in key features such as **deaths_civilians**, **country**, and **date_start** will be dropped. Missing values in non-critical features will be imputed using the column median. Categorical variables such as **country**, **type_of_violence**, **region**, and **side_b** will be encoded using one-hot encoding – converting each category into a binary column – to make them interpretable by machine learning models. Numerical features such as **latitude**, **longitude**, and **best** will be scaled using *StandardScaler* to ensure that no single feature dominates due to any differences in range. Supplementary features taken from ACLED and Airwars will be merged with the primary UCDP-GED dataset by matching on country, date, and the approximate location coordinates.

Before modeling, exploratory data analysis will be done to understand the distribution of civilian casualties, identify trends over time, examine potential patterns across presidential administrations, and surface any outliers or anomalies in the data. Visualisations will include histograms and bar charts for distributions and trends, boxplots grouped by important categorical variables, scatter plots for outlier detection, correlation heatmap for feature relationships, and spatial map for geographic clustering.

This project implements a progressive modeling pipeline, beginning with interpretable baseline models and advancing to ensemble methods. Chosen baseline models include the following: KNN, which predicts based on similarity to nearby events, Decision Tree for interpretability and feature insight, and Naive Bayes as a fast

probabilistic point of reference. More advanced models include SVM for finding the optimal boundaries between high and low casualty events, Random Forest for maintaining robustness through averaging across multiple trees, and XGBoost as the strongest expected performer on tabular data.

Models will be evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and R^2 . The primary metric will be RMSE as it penalises large errors, which prevents underpredicting that can lead to humanitarian organisations arriving under-resourced, legal bodies having incomplete evidence, and the human cost to be minimised. MAE and R^2 will be used as supplementary metrics, for an interpretable average error in actual civilian death counts and how much of the variance in civilian deaths the model actually explains, respectively. All models will be evaluated using k-fold cross-validation to ensure results generalise beyond the training data.

5. Expected Contributions:

This project contributes to published research in two ways. First, this project produces a filtered, analysis-ready dataset of US and US-led coalition military strikes and their documented civilian casualties—a resource that does not currently exist in this form. Secondly, it establishes a systematic benchmark comparing multiple regression models on this task, thus providing a baseline performance of figures that future researchers can build upon. Beyond research, this has direct real-world implications. Humanitarian organisations can use the resulting model to pre-position aid in times of a US-led intervention. Legal bodies such as the ICC will have access to independently verified evidence free of governmental biases. Most importantly, the populations whose deaths have been systematically undercounted, ignored, and deprived of justice stand to benefit from a framework that documents and models the true human cost of US military action.

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